Extension of the RSA Production Model

Mechanism-level exploration and the 2×2 tradeoff

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Branch `feat/extension-v5` • dc subset ($N = 3196$) • 16 commits

Where we start: the paper-reported model

Incremental recursive speaker, hierarchical per-participant α offsets. **Theory-driven**: minimal parameters with clear motivation.

The 2×2 story (memo §1)

Model	ELPD LOO	Δ ELPD	dSE	Δ/dSE	p
Inc. recursive	-7180	0	—	—	—
Inc. static	-7199	-19	1.84	-10.28	$< .001$
Global static	-7626	-446	40.27	-11.08	$< .001$
Global recursive	-7628	-448	40.69	-11.02	$< .001$

- **Speaker architecture dominates**: inc $\gg$ glb (Δ/dSE ≈ 11).
- **Semantic regime within incremental**: recursive $>$ static (Δ/dSE $= 10.28$).
- **Semantic regime within global**: null.

But: the fit is poor

- $R^2 = 0.346$ across the 90 (condition $\times$ utterance) cells.
- **L1 aggregate residual** $= 3.17$ (cells with emp $\geq 2\%$).
- No lapse and no length bias in the reported model — γ and ε *hardcoded to* 0. All mass must come from the RSA softmax alone.

Seven systematic misfits (memo §5.3)

Condition	Sharpness	Utterance	Emp	Model	Δ
colour-sufficient	high	C	.716	.403	**-.313**
size-sufficient	low	SF	.365	.081	**-.284**
colour-sufficient	low	C	.628	.402	**-.225**
both-necessary	low	C	.009	.222	**+.214**
size-sufficient	high	SFC	.022	.211	**+.189**
both-necessary	high	C	.000	.175	**+.175**
size-sufficient	low	SFC	.032	.187	**+.155**

Diagnosis (memo §5.3)

Residuals decompose along two independent axes.

1. Context-dependence of 1-word C

- Under-predicted by $\sim$ .25–.31 in colour-sufficient
- Over-predicted by $\sim$ .18–.21 in both-necessary
- No single α_C can fix both.

2. Length residuals: saturation, not level shift

- 2-word (SF) under-predicted by $\sim$ .28 in blurred size-sufficient
- 3-word (SFC) over-predicted by $\sim$ .17 in size-sufficient
- No monotone change to a linear γ fixes both.

These diagnostics motivate the mechanism track.

Extension timeline (dc subset, $N = 3196$)

`erdc`, `zrdc`, `brdc` — size × colour distinguishing, form redundant.
Three `relevant_property` values inside: `first` (size sufficient), `both`, `second` (colour sufficient).

Step	Mechanism	New parameters	Literature cue
ext-v1	per-dim α, γ, ε	$\alpha_D, \alpha_C, \alpha_F, \gamma, \varepsilon$	dim-salience asymmetry
v5a	condition-gated colour boost	λ_C (step-0 only)	Pechmann 89; Rubio-Fernández 16
v5b	saturating length bias	γ_1, γ_2	overspec not monotone
v5 (F1)	+ csuff length offsets	$\delta\gamma_1, \delta\gamma_2$	context-dep. length bias
v5 (C1)	+ canonical-order penalty	μ_{nc}	Sproat 91; Cinque 94; Scott 02
v5 (F2)	+ sharpness-dependent length	η_1, η_2	overspec > under blur

Flag correction along the way: `COLOUR_SUFFICIENT_CONDITIONS` was first ($ercf$, $zrcf$, $brcf$); corrected to ($ercf$, $zrdc$) — only trials where colour alone identifies the target.

Stage 1: ext-v1 (already in memo §4.1)

Per-dimension α + explicit γ + lapse ε.

	ELPD	vs baseline
baseline	-7157	0
ext-v1	-6387	$+770$

	R^2	L1
baseline	.35	3.17
ext-v1	.60	3.17

Posterior: $\alpha_D = 6.92$, $\alpha_C = 1.85$, $\alpha_F = 1.89$, $\gamma = 2.31$, $\varepsilon = 0.23$.

But: seven memo §5.3 misfits persist; $\varepsilon = 0.23$ uncomfortably high.

Rejected pre-v5 extensions (memo §4.2–4.4)

Before v5, three condition-independent biases were tested. **All rejected**:

Variant	Mechanism	R^2	ELPD	Outcome
v2 (first-word intercepts)	ϕ_C, ϕ_F on the chosen utterance type's first adjective (S = reference)	.62	−6337	**Wrong sign.** $\phi_C = -6.89$ (penalty) instead of boost. Misfit is **context-dependent**; condition-independent intercepts can't capture "C up in csuff, C down in both-necessary".
v3 (step-level $\boldsymbol{\mu}$ w/ per-dim α)	logits $= \alpha_{\text{vec}} \cdot \log L_{\text{ref}} + \boldsymbol{\mu}_{\text{vec}}$	—	—	**Non-identifiable.** $n_{\text{eff}} \approx 2$, $\hat{R}$ up to 14.7 for μ_C. α and μ trade off on the same logit scale. Fundamental, not a sampling fix.
v4 (step-level $\boldsymbol{\mu}$ w/ single α)	Dropped per-dim α; kept μ_C, μ_F to avoid v3's identifiability	.54	−6449	**Worse than ext-v1.** Losing per-dim α hurt more than μ helped. Per-dim α captures something μ cannot.

Lesson for v5: the misfit is *context-dependent*, not a global utterance-type tilt. This motivated the **condition-gated** λ_C in v5a.

Stage 2a: λ_C alone (v5a)

Step-0 logit boost for **C** when `is_colour_sufficient` $= 1$.

	ELPD	Δ vs ext-v1	λ_C posterior
ext-v1	-6387	0	—
v5a	-6265	$+122$	3.4 (95% CI 2.8–4.0)

- Strong positive λ_C: colour gets extra first-step pull when it suffices.
- `second/sharp C`: emp .716, was .403, now $\sim$ .55.
- **CF still too high** in `second`: boost spreads to C-initial compounds.

Stage 2b: saturating γ alone (v5b)

Replace linear $\gamma \cdot k$ with $\gamma_1 \, \mathbb{I}[k \geq 1] + \gamma_2 \, \mathbb{I}[k \geq 2]$.

	ELPD	Δ vs ext-v1
ext-v1	-6387	0
v5b	-6381	$+6$

Essentially no improvement without λ_C.

The two mechanisms are **synergistic**: γ-saturation only matters after C-initial mass is correctly apportioned.

Stage 3: F1 — condition-dependent length (v5 + $\delta\gamma$)

Let the length bias **shrink** on colour-sufficient trials:
$\gamma_{1,\mathrm{eff}} = \gamma_1 + \delta\gamma_1 \cdot$ `is_csuff`.

	ELPD	Δ vs ext-v1
ext-v1	-6387	0
v5a	-6265	$+122$
v5 (F1)	-5933	$+454$

Posteriors: $\lambda_C = 5.0$, $\gamma_1 = 2.56$, $\gamma_2 = 0.92$, $\delta\gamma_1 = -1.89$, $\delta\gamma_2 = -3.51$, $\varepsilon = 0.16$.

Effective γ on csuff trials: 0.67, -2.59 — 3-word heavily penalised, short utterances favoured.

$R^2 = .67$, L1 $= 2.70$. **Remaining puzzle:** over-predicts `DC` and `DFC`, under-predicts `DCF`.

Stage 4: C1 — canonical-ordering penalty

Add $\mu_{\mathrm{nc}} \cdot \mathbb{I}[\text{utterance violates } D < C < F]$.
Non-canonical set: `CD`, `CDF`, `CFD`, `DFC`, `FD`, `FDC`, `FC`, `FCD`.

	ELPD	Δ vs ext-v1
v5 (F1)	−5933	+454
v5 (C1)	−5323	+1063

$\mu_{\mathrm{nc}} = -5.07$ — very strong canonical-English preference.
ε drops to **0.10**.

$R^2 = .91$, L1 $= 1.34$.

7 of the 8 user-flagged residuals fixed: `DCF`, `DF`, `DC`, `DFC`, `C`.
Side-effect: on `first/sharp`, bare `D` under-predicted (.082 vs emp .269) and `DF` over-predicted.

Stage 5: F2 — sharpness-dependent length (final)

Add η_1, η_2 offsets active on `is_sharp` trials.

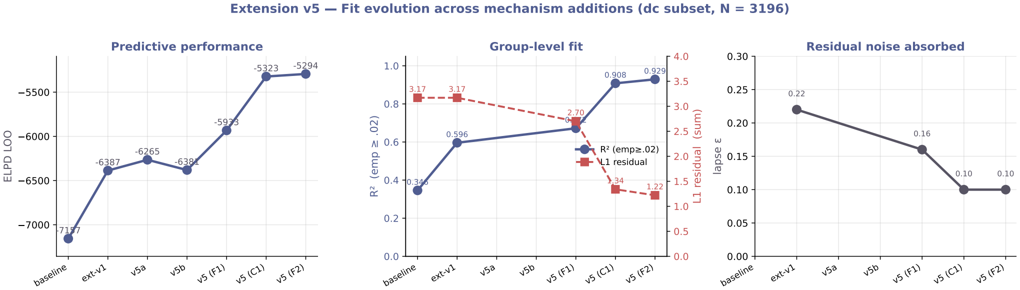
	ELPD	Δ vs ext-v1	R^2
v5 (C1)	-5323	$+1063$	.91
v5 (F2)	-5294	$+1093$	.94

Posteriors: $\eta_1 = -0.61$, $\eta_2 = -0.34$ (both informatively negative).
Effective γ on **sharp**: $1.71, 1.24$ vs on blurred: $2.32, 1.58$.

ε stays at 0.10; 0 divergences; all $\hat{R} \leq 1.02$.

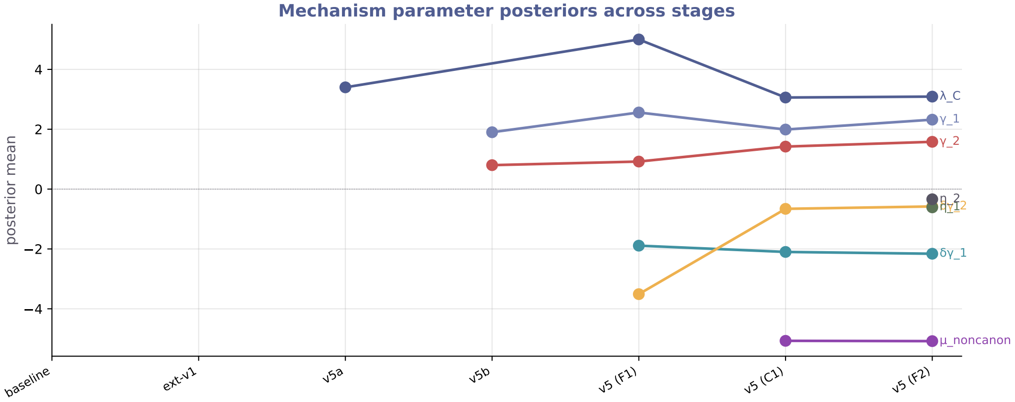
Stubborn residual: `first/sharp D` still under-predicted (.269 $\rightarrow$ .102).
Likely strategy heterogeneity: some participants minimum-describe, others hedge.

Fit evolution across stages



Each new mechanism shifts LOO and R^2 upward and ε downward. v5b (sat γ alone) flat vs ext-v1.

Mechanism posteriors across stages



Each parameter enters with a clear posterior sign and settles by F2. λ_C, μ_{nc} strong; η modest; $\delta\gamma_1$ strongly negative on csuff; $\delta\gamma_2$ shrinks once μ_{nc} absorbs canonical-order variance.

Full LOO & fit record — all stages

Stage	ELPD LOO	Δ vs baseline	Δ vs ext-v1	R^2	L1	ε
baseline	−7157	0	—	.35	3.17	—
ext-v1	−6387	+770	0	.60	3.17	.23
v2 (first-word intercepts) — rejected	−6337	+820	+50	.62	—	—
v3 (step-level μ, per-dim α) — rejected	non-identifiable ($\hat{R} \leq 14.7$)					
v4 (step-level μ, single α) — rejected	−6449	+708	−62	.54	—	—
v5a (λ_C only)	−6265	+892	+122	—	—	—
v5b (sat. γ only)	−6381	+776	+6	—	—	—
v5 (F1: $\lambda_C + \delta\gamma$)	−5933	+1224	+454	.67	2.70	.16
v5 (C1: + μ_{nc})	−5323	+1834	+1063	.91	1.34	.10
v5 (F2: + η)	−5294	+1863	+1093	.94	1.22	.10
v5_no_lm (F3: F2 with $\beta = 0$)	−5314	+1843	+1073	.92	1.30	.10

- LOO, R^2 (emp $\geq .02$), L1 (sum of $|\Delta|$ over 37 cells), ε lapse posterior mean.
- **Baseline has no lapse and no length bias** — γ and ε are *hardcoded to* 0 in `likelihood_function_reported_hier`, not sampled. All columns marked "—" are structurally absent, not just unreported.
- **v2–v4**: rejected exploratory variants from memo §4 — see previous slide for reasons.
- v5a and v5b: isolated mechanism probes (no full R^2/L1 reported in memo).
- Every add-on moves metrics monotonically in the right direction except v5b (sat γ alone, null) and v4 (which made things worse than ext-v1).

Ablation: is the LM prior still needed under F2?

Test: fit v5 (F2) with $\beta = 0$, effectively removing the LM prior. Every other mechanism identical.

Model	ELPD	Δ vs F2	R^2	L1
v5 (F2)	−5294	0	.928	1.22
v5_no_lm (F3)	−5314	−20.3	.915	1.30

Posterior compensation when LM is removed:

- λ_C: 3.09 $\rightarrow$ 3.33 (colour-boost works harder)
- γ_2: 1.58 $\rightarrow$ 1.32 (some 3-word preference re-absorbed)
- $\delta\gamma_1$: $-2.16 \rightarrow -2.39$ (more negative on csuff)
- $\mu_{\text{nc}}, \gamma_1, \eta, \varepsilon$: essentially unchanged.

Interpretation. LM prior contributes $\sim$20 ELPD of residual signal the explicit mechanisms don't absorb. Likely encoding frequency structure beyond canonical ordering + length (e.g., relative frequency among same-shape canonical utterances). Getting a non-zero gap is actually reassuring — it means $\mu_{\text{nc}} + \gamma + \eta$ are capturing structure *distinct* from the LM prior, not just refitting it.

Full parameter record — posterior means across stages

Parameter	baseline	ext-v1	v2	v4	v5a	v5b	v5 (F1)	v5 (C1)	v5 (F2)
α_D	single α	6.92	~ 6.9	single α	~ 6.9	~ 6.9	hier.	hier.	hier.
α_C		1.85	~ 1.8		~ 1.8	~ 1.9	hier.	hier.	hier.
α_F		1.89	~ 1.9		~ 1.9	~ 1.9	hier.	hier.	hier.
ϕ_C (first-word)	—	—	-6.89	—	—	—	—	—	—
ϕ_F (first-word)	—	—	-5.90	—	—	—	—	—	—
μ_C (step-level)	—	—	—	fitted	—	—	—	—	—
μ_F (step-level)	—	—	—	fitted	—	—	—	—	—
λ_C	—	—	—	—	$+3.4$	—	$+5.0$	$+3.06$	$+3.09$
γ (linear)	—	2.31	~ 2	~ 2	—	—	—	—	—
γ_1	—	—	—	—	—	$+1.9$	$+2.56$	$+1.99$	$+2.32$
γ_2	—	—	—	—	—	$+0.8$	$+0.92$	$+1.42$	$+1.58$
$\delta\gamma_1$	—	—	—	—	—	—	-1.89	-2.10	-2.16
$\delta\gamma_2$	—	—	—	—	—	—	-3.51	-0.66	-0.58
η_1	—	—	—	—	—	—	—	—	-0.61
η_2	—	—	—	—	—	—	—	—	-0.34
μ_{nc}	—	—	—	—	—	—	—	-5.07	-5.08
$\log \beta$	~ 2	2.05	~ 2	~ 2	~ 2	~ 2	2.05	-0.18	-0.17
ε	—	0.23	$\sim .2$	$\sim .2$	—	—	0.16	0.10	0.10
τ	—	0.92	—	—	—	—	0.94	0.60	0.62

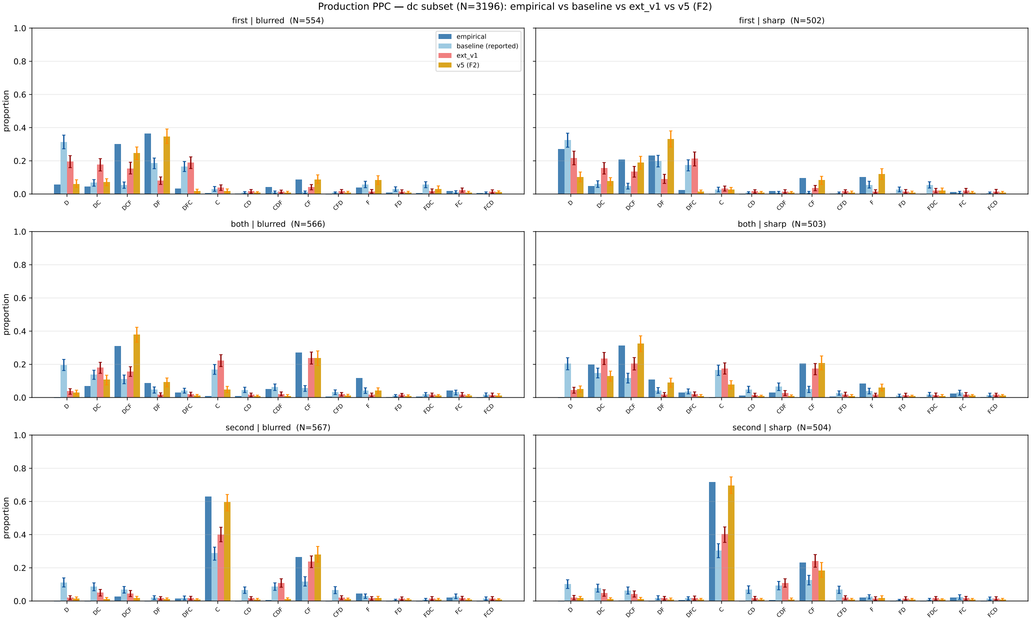
v3 omitted (non-identifiable). v2 / v4 columns show posterior means as reported in memo §4.2 / §4.4.

- "hier." $= \alpha$ sampled hierarchically with per-participant δ_p; population means close to ext-v1's values.

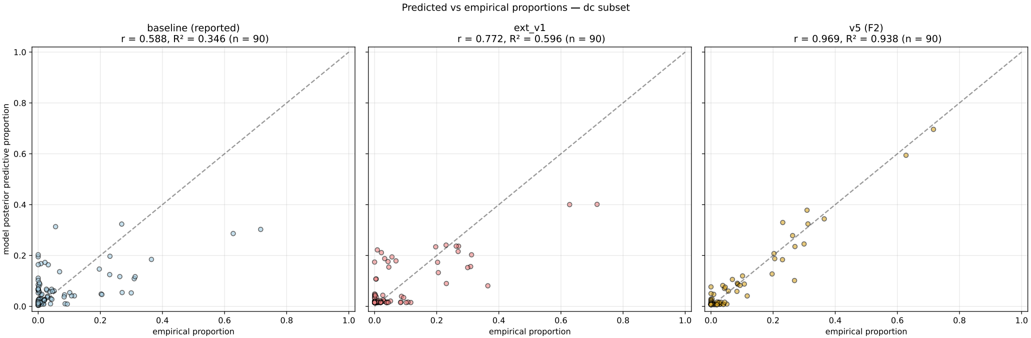
Final v5 (F2) parameters — full record

Parameter	Prior	Posterior mean (95% CI)	$\hat{R}$
α_D	HalfN(5)	(hier. w/ δ_p)	≤ 1.01
α_C	HalfN(5)	(hier.)	≤ 1.01
α_F	HalfN(5)	(hier.)	≤ 1.01
λ_C	N(0,1)	$+3.09$ (2.67, 3.55)	1.00
γ_1	N(0,1)	$+2.32$ (2.16, 2.49)	1.00
γ_2	N(0,1)	$+1.58$ (1.43, 1.74)	1.01
$\delta\gamma_1$	N(0,1)	-2.16 (-2.33, -1.97)	1.00
$\delta\gamma_2$	N(0,1)	-0.58 (-1.21, $+0.04$)	1.00
η_1	N(0,1)	-0.61 (-0.77, -0.43)	1.00
η_2	N(0,1)	-0.34 (-0.51, -0.15)	1.00
μ_{nc}	N(0,1)	-5.08 (-5.61, -4.56)	1.01
$\log\beta$	N(0,0.5)	-0.17	1.00
ε	Beta(1,50)	0.10 (0.09, 0.12)	1.00
τ	HalfN(0.2)	0.62 (0.51, 0.73)	1.00

PPC barplot — baseline vs ext-v1 vs v5 (F2)



Predicted vs empirical ($n = 90$ cells)



R^2: baseline .35 $\rightarrow$ ext-v1 .60 $\rightarrow$ **v5 (F2)** .94.

The 2×2 question: does the paper story survive?

Under F2 mechanisms we re-ran the 2×2 (speaker architecture $\times$ semantics regime) on the dc subset.

Three parameterisations to compare:

1. **Paper-reported (memo §1)**: minimal, theory-driven.
2. **F2-fixed globals**: global cells use $\gamma/\delta\gamma/\eta/\mu_{\mathrm{nc}}$ fixed at inc-recursive posterior means.
3. **F2-full globals**: every parameter sampled freely in every cell. Fairest comparison.

2×2: paper-reported

	recursive	static	Δ(rec−sta)
incremental	−7180	−7199	+19 ($\Delta/\mathrm{dSE} = 10.28$)
global	−7628	−7626	−2 (null)
Δ(inc−glb)	+448	+427	
Δ/dSE	11.02	11.08	

Paper narrative

- Architecture dominates: inc $\gg$ glb, $\Delta/\mathrm{dSE} \approx 11$.
- Recursive helps incremental a lot, does not help global.
- Diff-in-diff $= +21$ (inc benefits more from rec).

2×2: F2 with globals' params fixed at inc posterior means

	recursive	static	Δ(rec−sta)
incremental	-5294	-5297	$+2.80$ ($\Delta/\mathrm{dSE} = 2.22$, marginal)
global (fixed)	-5600	-5619	$+19.15$ ($\Delta/\mathrm{dSE} = 6.99$)
Δ(inc−glb)	$+306$	$+323$	
Δ/dSE	10.51	11.03	

- Architecture still dominant ($\Delta/\mathrm{dSE} \approx 10.5$).
- **Interaction has flipped**: rec now helps global, not incremental.
- But global's length/order parameters are fixed at inc values — unfair test.

2×2: F2 with globals fully sampled — the fair picture

	recursive	static	Δ(rec$-$sta)
incremental	-5294	-5297	$+2.80$ (Δ/dSE $= 2.22$)
global (full)	-5357	-5365	$+8.25$ (Δ/dSE $= 3.82$)
Δ(inc$-$glb)	$+63$	$+69$	
Δ/dSE	3.18	3.48	

Key changes vs paper-reported

- Architecture effect Δ/dSE: $11 \to 3.2$ ($\sim 1/3$ **of paper's**).
- Regime-within-inc Δ/dSE: $10.28 \to 2.22$ ($\sim 1/5$ **of paper's**).
- Regime-within-glb: $\approx 0 \to 3.82$ (appears, modestly).
- Diff-in-diff: $+21 \to +5.45$ (same sign, small magnitude).

Full record — side-by-side

	Paper	F2-fixed	**F2-full**
R^2 (all cells)	.35	.94	.94
L1 residual	3.17	1.22	1.22
Lapse ε	0.22	0.10	0.10
Arch effect Δ/dSE	± 11	± 10.5	± 3.2
Regime |inc Δ/dSE	10.28	2.22	2.22
Regime |glb Δ/dSE	≈ 0	6.99	3.82
Interaction (diff-in-diff)	$+21$	-16	$+5.45$
Rec helps...	incremental	global (artefact)	slightly global

The tradeoff

Theory track (paper)

- Minimal, motivated mechanisms
- Clear 2×2 interaction
- **Poor group-level fit ($R^2 = .35$)**
- **Seven large misfits in §5.3**
- Lapse absorbs unexplained structure

Mechanism track (F2)

- Rich; each term literature-grounded
- Excellent fit ($R^2 = .94$)
- All seven misfits resolved
- Lapse halved ($0.22 \to 0.10$)
- **Architecture signal shrinks to $\Delta/\text{dSE} = 3$**
- **Regime-within-inc weakens to marginal**

Core tension. Mechanisms designed to reduce residuals explain variance that the paper's minimal parameterisation attributed to **architecture** $\times$ **regime**. The interaction survives in reduced form.

Two readings

Reading I: mechanisms as rivals
The paper's interaction was partly *architectural sparsity fitting error structure*. Once error is absorbed by explicit mechanisms (colour salience, canonical ordering, context-dep. length), the residual architectural signal is modest. The main claim must weaken.

Reading II: mechanisms as scaffolding
λ_C, μ_{nc}, $\delta\gamma$, η are *implementations* of what context-updating $\times$ incremental architecture is *supposed* to do. Of course adding them absorbs architectural signal — they are the phenomena the architecture was indirectly capturing.

Both are defensible. The choice shapes the paper's framing.

Four concrete decisions

1. **Report v5 (F2) or keep the paper-reported model?**
   A middle path: main-text keeps the reported model; supplementary section shows F2 as mechanism-level refinement.
2. **If we adopt F2, how do we frame the interaction?**
   Reading I softens; Reading II preserves. A referee will ask.
3. **Adopt the corrected `COLOUR_SUFFICIENT` flag ($ercf$, $zrdc$)?**
   Independent of the F2 decision. Probably yes — the definition is simply more accurate.
4. **Generalise beyond the dc subset?**
   Parallel runs on df (size+form) and cf (colour+form) would test whether F2 mechanisms are symmetric or colour-specific.

Open questions

- **Strategy heterogeneity.** $\varepsilon = 0.10$ is smaller than ext-v1's 0.22 but non-zero. The `first/sharp` bare-`D` vs `DF` split looks like two participant sub-populations.
- **Full-data ($N = 9586$) run.** Our early full-data run used the incorrect flag. A clean full-data F2 run would take $\sim$15 min on GPU.
- **df and cf subsets.** Do the mechanisms generalise? If a "form-sufficient" flag is needed, symmetric structure; if not, colour-specificity.
- **LOO warnings.** `az.compare` flagged Pareto $k > 0.7$ in some cells. Differences reliable in direction, noisier in magnitude.

Summary

- Paper-reported model: theory-driven, interpretable 2×2, but $R^2 = .35$.
- Residuals trace to three independent phenomena: **colour salience**, **canonical ordering**, **condition-dependent length bias**.
- Adding all three (v5 F2) lifts fit to $R^2 = .94$ and lapse to .10, with every new parameter informative, 0 divergences.
- Re-running the 2×2 under F2 with **freely sampled** globals: architecture effect shrinks $\sim 3\times$; regime effect shrinks and partially flips direction.
- Two readings of the tradeoff; both defensible.
- Decision the paper needs: **report F2, report paper-reported, or both?**

Artefacts: branch `feat/extension-v5`, spec at `docs/superpowers/specs/...`, synthesis at `10-writing/extension_v5_synthesis.md`.

Thanks

Questions and discussion.

`feat/extension-v5`	16 commits
`10-writing/extension_v5_synthesis.md`	detailed write-up
`10-writing/extension_v5_slides_csp.tex`	this deck
`05-modelling-production-data/figures/v5/`	figures used above
`05-modelling-production-data/results/`	LOO + residual CSVs